

Affordance-Grasp Conditioned Spill-Free Trajectory Optimization

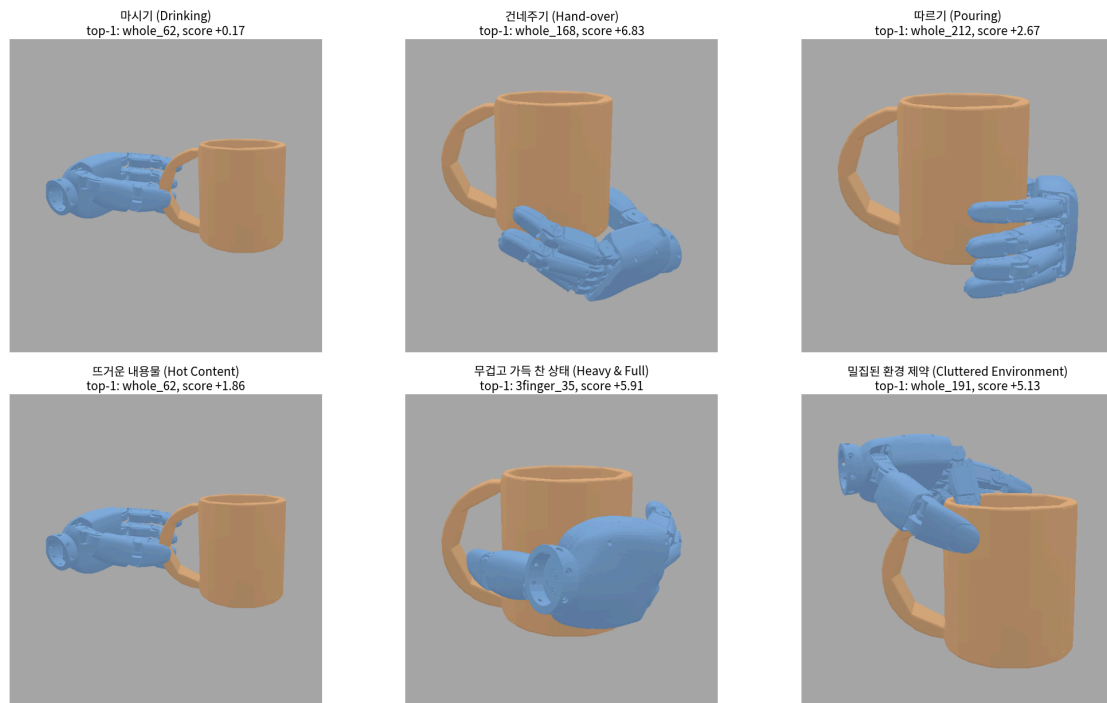
Does **how you grasp** a cup change **how safely you can carry** it?

Motion Planning · Proposal

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Context — my thesis: task-conditional grasp generation

Stage 1 Best-of-N — same 900 mug grasps, reward chooses per task



- **Input:** object mesh + a natural-language **task**
- **Output:** dexterous (multi-finger) **grasp pose** — which fingers, where on the object
- Same mug, **task changes** → **grasp changes** (affordance / grasp taxonomy)

Same 900 mug grasps; the task selects a different one. (my thesis, Stage 1)

This project starts where the grasp ends → the *carry*.

Problem — carry liquid from A to B without spilling

Given a grasped mug, plan a 6-DoF end-effector trajectory that is:

- **(a)** spill-free **(b)** obstacle-avoiding **(c)** smooth **(d)** tilt $\leq \theta_{max}$

Even a perfectly upright cup spills under a fast stop or turn — the danger is *motion*, not just orientation.

Why? That is the next slide.